# Malaviya National Institute of Technology Jaipur

## Department of Mathematics

**B.Tech. First Semester | Odd Semester - 2025-26**

**Mid-Term Examination, September - 2025**

**22MAT101: MATHEMATICS - I**

**Duration:** 90 Minutes **Max Marks:** 20

*(Answer all the questions. All questions carry equal marks. Use of calculator is not allowed.)*

1. Define rank of a matrix. Find the rank of the following matrix by converting it to column echelon form.

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix}.$$

2. Determine the values of $\lambda$ for which the following system of linear equations is consistent. Hence, solve the system in each case.

$$x + y + z = 1,$$
$$2x + y + 4z = \lambda,$$
$$4x + y + 10z = \lambda^2.$$

3. Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & -1 \\ 1 & 1 & 0 \end{bmatrix}$. Is the matrix $A$ diagonalizable? If yes, diagonalize it. If no, justify your answer.

4. (a) Let $A$, $B$, and $P$ be square matrices of order $n$ with $B = P^{-1}AP$. Suppose $\mathbf{x}$ is an eigenvector of $A$ corresponding to the eigenvalue $\lambda$. Prove that $\lambda$ is also an eigenvalue of $B$, and determine a corresponding eigenvector of $B$. **[2]**

(b) Let $f = f(u, v, w)$ with $u = x + y + z$, $v = xy + yz + zx$, and $w = xyz$. Show that
$$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} = u\frac{\partial f}{\partial u} + 2v\frac{\partial f}{\partial v} + 3w\frac{\partial f}{\partial w}.$$ **[2]**

5. Consider the function

$$f(x, y) = \begin{cases} \dfrac{x^3 y}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0), \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

$y\,\frac{x^3}{x^2+y^2}$

(a) Show that the function is continuous at the origin.

(b) Show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$.

$\frac{\partial f}{\partial x}$ $\frac{\partial f}{\partial y}$

$y\left[\frac{(x^2+y^2)(3x^2) - (x^3)(2x)}{(x^2+y^2)^2}\right]$

$6x^2y(x^4+y^4+2x^2y^2)$

$y\left\{\frac{x^4+3x^2y^2}{\ldots}\right\} = \frac{x^4+3x^2y^2}{(x^2+y^2)^2} + y\left\{\frac{(x^2+y^2)^2((3x^2)(2y)) - (x^4+3x^2y^2)\,2(x^2+y^2)(\ldots)}{(x^2+y^2)^4}\right.$ $4y(x^2 \ldots$

22MAT101- Mathematics-I
Mid-Term Examination 2024-25 (Odd Semester)
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**Total Marks: 30** **Duration-90 Minutes** **Date: 23-09-2024**

**All questions are compulsory.**
**Calculators are NOT allowed.**

1. Find the values of $\lambda$ and $\mu$ for which the following system of linear equations has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions. Also, find the solution in the case of infinitely many solutions.

$$x + y + z = 6, \qquad 2x + 5y + \lambda z = \mu, \qquad x + 2y + 3z = 14.$$

2. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, then show that $A^n = A^{n-2} + A^2 - I$ for $n \geq 3$. Hence, find $A^{50}$.

3. (a) Show that similar matrices have same eigen values. [**2** marks]

   (b) Let $A$ be a diagonalizable matrix. Show that the transpose of $A$, that is, $A^t$ is also diagonalizable. [**3** marks]

4. (a) Show that the limit $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2 y}{x^4 + y^2}$ does not exist. [**2** marks]

   (b) Discuss the continuity of the following function at the point $(0, 0)$ [**3** marks]

$$f(x, y) = \begin{cases} \dfrac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

5. Transform the following equation into polar coordinates

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

6. If $u(x, y) = \sin^{-1} \dfrac{x^2 + y^2}{x + y}$, then show that

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \tan u$$

and

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = \tan^3 u.$$

[**2+3** marks]